NOETIC-STYLE · GRADE 4 · FREE DIAGNOSTIC

Noetic-Style Grade 4 Diagnostic Test

20	50 min	No
questions	time	calculator

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1. 20 problems, easy to hard, at real contest length. Write each answer in its box.
2. Read each problem twice and circle what it asks for. A blank box scores zero, so make your best try.
3. Stuck? Skip it and come back. After the test, scan the code next to any problem you missed to watch the key move for it.

Name _____ Date _____ Score _____ / 20

1 Leo's robot vacuum cleans 18 square feet of floor every 3 minutes. The playroom floor is 120 square feet. How many minutes does the robot need to clean the whole playroom?

Stuck? Scan for the key

Answer ☐

2 A soccer practice starts at 3:45 pm and ends at 5:20 pm. How many minutes long is the practice?

Stuck? Scan for the key

Answer ☐

3 At the science fair, Priya puts 3 identical clay balls and a 5-gram weight on the left pan of a scale. On the right pan she puts 2 more of the clay balls and a 17-gram weight. The pans are level. How many grams does one clay ball weigh?

Stuck? Scan for the key

Answer ☐

4 4 pencils cost the same as 1 notebook. 3 notebooks cost the same as 2 rulers. How many pencils cost the same as 1 ruler?

Stuck? Scan for the key

Answer ____

5 A tall candle is 18 cm long and a short candle is 11 cm long. Both are lit at the same time, and each burns down 1 cm every hour. After how many hours will the tall candle be exactly twice as long as the short one?

Stuck? Scan for the key

Answer ____

6 A class wants to collect 250 cans for a food drive. On Monday they collect 64 cans and on Tuesday they collect 87 cans. How many more cans do they still need?

Stuck? Scan for the key

Answer ____

7 A farm has 3 times as many hens as ducks. There are 56 more hens than ducks. How many birds (hens and ducks) are on the farm?

Stuck? Scan for the key

Answer ____

8 A red bag holds cards numbered 1, 2, 3, 4 and 5. A blue bag holds cards numbered 6, 7, 8 and 9. Ana picks one card from each bag. How many different pairs of cards have an odd sum?

Stuck?
Scan for the key

Answer ____

9 Nora's birthday is on a Saturday. Her cousin's birthday is 60 days later. On what day of the week is her cousin's birthday?

Stuck?
Scan for the key

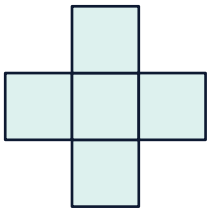
Answer ____

10 Kim had some marbles. She won 9 more at recess. Then she gave half of all her marbles to her brother, and later she lost 4. Now she has 13. How many marbles did Kim have at first?

Stuck?
Scan for the key

Answer ____

11 Five equal squares form the plus sign shown. Each square has a perimeter of 12 cm. What is the perimeter of the plus sign?



Stuck?
Scan for the key

Answer ____

12 Mr. Lee glues 24 one-inch cubes into a solid block 2 cubes deep, 3 cubes wide and 4 cubes tall, then dips the whole block in blue paint. When he pulls the cubes apart, how many show blue on exactly 2 faces?

Stuck? Scan for the key

Answer ____

13 Amy visits the library every 4 days and Ben visits every 6 days. They both visited today. In the next 60 days, how many more times will they visit on the same day?

Stuck? Scan for the key

Answer ____

14 At a fair, kid tickets cost $3 and adult tickets cost $5. A group paid exactly $38 for their tickets. How many different mixes of kid and adult tickets could the group have bought?

Stuck? Scan for the key

Answer ____

15 Of the 32 kids at a sleepover, 6 ate no popcorn and no pretzels. 19 kids ate popcorn and 14 kids ate pretzels. How many kids ate popcorn but no pretzels?

Stuck? Scan for the key

Answer ____

16 Sam cuts a long log into 7 pieces. Each cut takes 4 minutes, and after every cut except the last one he rests for 1 minute. How many minutes does the whole job take?

Stuck? Scan for the key

Answer ____

17 A soccer team played 5 games and scored 34 goals in all. It scored a different number of goals in every game, but never more than 9 goals in one game. In its worst game, what is the fewest goals it could have scored?

Stuck? Scan for the key

Answer ____

18 Three fish tanks hold 72 fish in all. A worker moves 5 fish from tank A to tank B, then 8 fish from tank B to tank C, then 3 fish from tank C to tank A. Now the three tanks hold the same number of fish. How many fish were in tank B at first?

Stuck? Scan for the key

Answer ____

19 Ana, Ben, Cara and Dev each have a different pet: a cat, a dog, a fish and a bird. Ana's pet has no fur. Ben's pet cannot fly and cannot swim. Cara's pet is not the bird. Dev has the dog. Who has the fish?

Stuck? Scan for the key

Answer ____

20 Gym lockers have 3-digit codes made from the digits 1, 2, 3, 4 and 5, with no digit repeated. A code must be odd and larger than 300. How many such codes are there?

Stuck? Scan for the key

Answer ____

End of test. Check that every box has an answer.

NOETIC-STYLE · GRADE 4 · ANSWER KEY

Diagnostic Test: Answers

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Each problem uses one tool from the channel. For any you missed, scan the code next to that problem on the test to watch the lesson, then try the problem again tomorrow without looking.

1 20 Tool #10: One Unit First

One minute first: 18 ÷ 3 = 6 square feet a minute. Then 120 ÷ 6 = 20 minutes.

2 95 Tool #47: Jump to the Hour

Jump to the hour: 3:45 to 4:00 is 15 min, 4:00 to 5:00 is 60, 5:00 to 5:20 is 20. 15 + 60 + 20 = 95.

3 12 Tool #48: Take the Same Off Both Sides

Take 2 clay balls and 5 grams off both pans: 1 ball balances 17 − 5 = 12 grams.

4 6 Tool #2: Swap It

3 notebooks = 12 pencils = 2 rulers. So 1 ruler = 12 ÷ 2 = 6 pencils.

5 4 Tool #17: Keep the Gap

The gap stays 18 − 11 = 7 cm. Twice as long means the gap equals the short candle: 7 cm. 11 − 7 = 4 hours.

6 99 Tool #1: Whole Minus Part

Whole minus parts: 250 − 64 − 87 = 99 cans still needed.

7 112 Tool #30: Smaller Is One Block

Ducks = 1 block, hens = 3 blocks. The 2 extra blocks are 56, so 1 block = 28. In all 4 blocks = 112.

8 10 Tool #49: Multiply the Choices

Odd sum = odd + even. Red odd (3) × blue even (2) = 6, plus red even (2) × blue odd (2) = 4. 6 + 4 = 10.

9 Wednesday Tool #5: Find the Cycle

60 = 8 weeks + 4 days. 8 weeks lands on Saturday again; 4 more days: Sunday, Monday, Tuesday, Wednesday.

10 25 Tool #39: Work Backwards

Undo from the end: 13 + 4 = 17, double it: 34, take off the 9 she won: 25.

11 36 Tool #37: Count the Sides

Each square's side is 12 ÷ 4 = 3 cm. Walk around the outside: 12 sides. 12 × 3 = 36 cm.

12 12 Tool #51: Sort the Cubes

Sort the 24 cubes. The block is 2 deep, so every cube is on the front or back. 8 corners have 3 faces, 4 middles have 1. 24 − 8 − 4 = 12.

13 5 Tool #52: Find the First Match

First match: the smallest number in both lists is 12. Then jump by 12: days 12, 24, 36, 48, 60. That is 5 times.

14 3 Tool #3: Test Before You Count

Test each adult count: 1 leaves 33 (11 kids), 4 leaves 18 (6 kids), 7 leaves 3 (1 kid). Others leave a number 3 does not divide: 3 mixes.

15 12 Tool #21: Neither First

Neither first: 32 − 6 = 26 ate a snack. Both = 19 + 14 − 26 = 7. Popcorn only = 19 − 7 = 12.

16 29 Tool #50: Act It Out

Act it out: 7 pieces need only 6 cuts (24 min) and 5 rests (5 min), since there is no rest after the last cut. 29 minutes.

17 4 Tool #4: Push to the Extreme

Push the other four games as high as allowed: 9 + 8 + 7 + 6 = 30. The worst game is at least 34 − 30 = 4.

18 27 Tool #20: Keep the Total

Moves keep the total: at the end each tank has 72 ÷ 3 = 24. Undo tank B's moves: 24 + 8 − 5 = 27.

19 Cara Tool #36: Logic Grid

Dev has the dog, so Ben (fur, no wings or fins) has the cat. Ana has no fur: fish or bird. Cara is not the bird, so Cara has the fish.

20 21 Tool #24: Pickiest Box First

Pickiest boxes: the first digit (3, 4 or 5) and the last (1, 3 or 5) share 3 and 5. First 4: 3 × 3 = 9. First 3 or 5: 2 × 2 × 3 = 12. 9 + 12 = 21.